

aPTT Mixing Studies

Test Details

METHODOLOGY

To perform mixing studies, the patient specimen is mixed 1:1 with normal platelet-free plasma and with veronal buffered saline. aPTTs are performed on these mixtures. An aliquot of the normal plasma/patient plasma mixture is also incubated at 37°C for one to two hours and then an aPTT is performed to further clarify the cause of the prolongation. Normal platelet-free plasma and patient plasma are also incubated separately and then mixed after incubation as a control for possible factor degradation during incubation.

RESULT TURNAROUND TIME

2 - 3 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Activated partial thromboplastin time (aPTT); aPTT 1:1 mix normal plasma (NP); aPTT 1:1 mix saline; aPTT 1:1 mix incubated; aPTT 1:1 mix NP, incubated control

Use

Distinguish specific (ie, factor VIII) and nonspecific (lupus anticoagulant) inhibitors; determine heparin presence; detect single or multiple factor deficiencies

Limitations

Lupus anticoagulant sensitive reagents are more responsive to the presence of lupus anticoagulants and are used in this panel. aPTT mixing studies will not be performed unless the aPTT is prolonged to 5 seconds or more above the upper end of the reference interval as they cannot be accurately interpreted when the aPTT is only slightly prolonged. The aPTT may not be extended in individual factor deficiencies unless the levels drop to 25% to 45% depending on the factor.⁶ Less significant individual factor deficiencies in combination (multiple factor deficiency) can extend the aPTT.⁶ The aPTT is more sensitive to intrinsic pathway factor deficiencies than common pathway factor deficiencies.⁶ Factor VIII elevations, as can occur due to acute phase reactions, can normalize a mildly extended aPTT result.⁶

Custom Additional Information

An isolated prolonged aPTT result with a normal prothrombin time implies either the presence of heparin and inhibitor (either specific factor or lupus anticoagulant) or a factor deficiency of the intrinsic system.⁷ Unless the cause for the prolonged aPTT is known, as in the case of heparin therapy, mixing studies may be necessary to determine the etiology of the prolonged result. If the result obtained with the immediate normal plasma mixture corrects to within the reference range and the saline mixture result increases dramatically, a factor deficiency or specific factor inhibitor is suspected.⁷

With factor deficiency, heparin, or a specific factor inhibitor, the 1:1 saline mix shows significant prolongation of the aPTT result. If the result obtained with the immediate normal plasma mixture shows only partial or no correction and the result obtained with the saline mixture shows near correction to the reference range or only mild prolongation, then a nonspecific inhibitor, such as lupus anticoagulant (LA), is suspected. After incubation at 37°C, an aPTT result equivalent to that of the original mixture is indicative of a factor deficiency or heparin. If the result of the aPTT on the incubated mixture demonstrates further prolongation (with the control remaining equivalent to the original result), a specific factor inhibitor is indicated. A specific factor VIII inhibitor may show initial correction of the original mixture, but will demonstrate prolongation upon incubation.

If mixing studies indicate the possibility of single or multiple factor deficiencies, a PT may be performed to differentiate between common pathway and intrinsic pathway deficiencies. Factor assays should then be performed to determine specific factors involved. If a factor inhibitor is suspected, specific factor assays and inhibitor titers may be necessary. If heparin is suspected without clinical confirmation, a thrombin time with addition of heparin neutralizer should be performed on the original specimen. Plasma suspected of containing a lupus anticoagulant (LA) should be further tested with low phospholipid tests and associated confirmatory tests. A patient with an initial positive antiphospholipid antibody or lupus anticoagulant test should be tested after 12 or more weeks to determine if the antibody is persistent.

Specimen Requirements

Specimen

Plasma, **frozen**

Volume

2 mL

Container

Blue-top (sodium citrate) tube

Collection Instructions

Citrated plasma samples should be collected by double centrifugation. Blood should be collected in a blue-top tube containing 3.2% buffered sodium citrate.¹ Evacuated collection tubes must be filled to completion to ensure a proper blood to anticoagulant ratio.^{2,3} The sample should be mixed immediately by gentle inversion at least six times to ensure adequate mixing of the anticoagulant with the blood. A discard tube is not required prior to collection of coagulation samples, except when using a winged blood collection device (ie, "butterfly"), in which case a discard tube should be used.^{4,5} When noncitrate tubes are collected for other tests, collect sterile and nonadditive (red-top) tubes prior to citrate (blue-top) tubes. Any tube containing an alternate anticoagulant should be collected after the blue-top tube. Gel-barrier tubes and serum tubes with clot initiators should also be collected after the citrate tubes. Centrifuge and carefully remove the plasma using a plastic transfer pipette, being careful not to disturb the cells. Deliver to a plastic pipette to remove the plasma, staying clear of the platelets at the bottom of the tube. Transfer the plasma into a Labcorp PP transpak frozen purple tube with screw cap (Labcorp No. 49482). Freeze immediately and maintain frozen until tested.

Please print and use the *Volume Guide for Coagulation Testing*

(</tests/related-documents/L5901>) to ensure proper draw volume.

Storage Instructions

Freeze.

Patient Preparation

Ideally, the patient should not be on anticoagulant therapy. Avoid warfarin (Coumadin®) therapy for two weeks prior to the test and heparin, direct Xa, and thrombin inhibitor therapies for about three days prior to testing. Do **not** draw from an arm with a heparin lock or heparinized catheter.

Causes for Rejection

Severe hemolysis; improper labeling; clotted specimen; specimen diluted with IV fluids; samples thawed in transit; improper sample type; sample out of stability

Footnotes

1. Adcock DM, Kressin DC, Marlar RA. Effect of 3.2% vs 3.8% sodium citrate concentration on routine coagulation testing. *Am J Clin Pathol*. 1997 Jan; 107(1):105-110. PubMed 8980376 (<http://www.ncbi.nlm.nih.gov/pubmed/8980376>)

2. Reneke J, Etzell J, Leslie S, Ng VL, Gottfried EL. Prolonged prothrombin time and activated partial thromboplastin time due to underfilled specimen tubes with 109 mmol/L (3.2%) citrate anticoagulant. *Am J Clin Pathol*. 1998 Jun; 109(6):754-757. PubMed 9620035 (<http://www.ncbi.nlm.nih.gov/pubmed/9620035>)

3. National Committee for Clinical Laboratory Standardization. *Collection, Transport, and Processing of Blood Specimens for Coagulation Testing and General Performance of Coagulation Assays; Approved Guideline*. 5th ed. Villanova, Pa: NCCLS; 2008. Document H21-A5:28(5).

4. Gottfried EL, Adachi MM. Prothrombin time and activated partial thromboplastin time can be performed on the first tube. *Am J Clin Pathol*. 1997 Jun; 107(6):681-683. PubMed 9169665 (<http://www.ncbi.nlm.nih.gov/pubmed/9169665>)

5. McGlasson DL, More L, Best HA, Norris WL, Doe RH, Ray H. Drawing specimens for coagulation testing: Is a second tube necessary? *Clin Lab Sci*. 1999 May-Jun; 12(3):137-139. PubMed 10539100 (<http://www.ncbi.nlm.nih.gov/pubmed/10539100>)

6. Van Cott EM, Laposata M. Coagulation. In Jacobs DS, DeMott WR, Oxley DK, eds. *Laboratory Test Handbook With Key Word Index*. Hudson, Ohio: Lexi-Comp; 2001: 327-358.

7. Adcock DM, Bethel MA, Macy PA. *Coagulation Handbook*.

(<http://www.esoterix.com/sites/default/files/imce/uploads/L3692.pdf>) Aurora, Colo:

Esoterix—Colorado Coagulation; 2006.